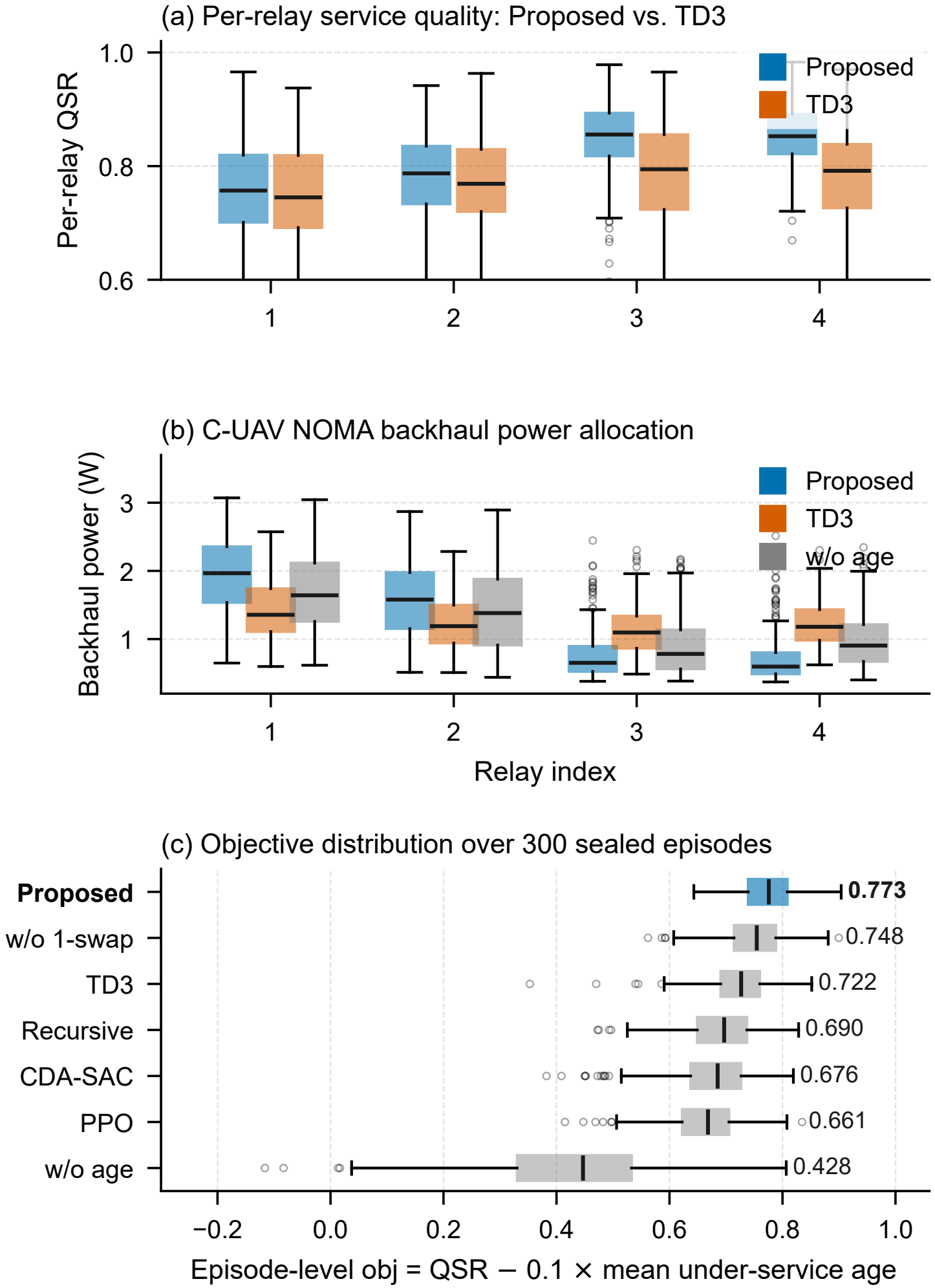

(a) Per-relay service quality: Proposed vs. TD3
Per-relay QSR
1.0
0.8
0.6
1
2
3
4
Proposed
TD3
(b) C-UAV NOMA backhaul power allocation
Backhaul power (W)
3
2
1
1
2
3
4
Relay index
Proposed
TD3
w/o age
(c) Objective distribution over 300 sealed episodes
Proposed
w/o 1-swap
TD3
Recursive
CDA-SAC
PPO
w/o age
0.773
0.748
0.722
0.690
0.676
0.661
0.428
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Episode-level obj = QSR − 0.1 × mean under-service age